

Reformulator



Autoformalizer



Problem: Find the smallest constant C such that for all real numbers x and y , the following inequality holds:

$$x^2 + x + y^2 + y + C \geq xy$$

Answer: $C = 1$

Original Problem

Proof problem: For all real numbers x and y , please prove:

$$x^2 + x + y^2 + y + 1 \geq xy$$

Proof Problem

```
import Mathlib.Data.Real.Basic

/-- For all real numbers x and y,
    x^2 + x + y^2 + y + 1 ≥ x*y. -/
theorem quad_ineq (x y : ℝ) : x^2 + x +
    y^2 + y + 1 ≥ x * y := by sorry
```

Lean4 Code